

Zirui Zhao

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No.28 Xianning West Road, Xi'an, China, 710049.

EDUCATION

School of Computing, National University of Singapore

Doctor of Philosophy in Computer Science

Singapore

Start in August 2020

Faculty of Electronic and Information Engineering, Xi'an Jiaotong University

Bachelor of Engineering in Automation Science and Technology (Honors Qian Xuesen Program)

Xi'an, China

Aug. 2016 - Jun. 2020

VISITING

Carnegie Mellon University

Work in the Safe AI Lab at Robotics Institute & Dept. Mechanical Engineering.

Pittsburgh, U.S.A.

Jul 2019 - Aug 2019

National University of Singapore

Participated in "Tele-Robotic and Deep Learning" Program at 2018 Summer Workshop

Singapore

Jul 2018 - Aug 2018

SKILLS

- **Languages:** : Python, C++, MATLAB. **Technologies:** GitHub, Carla.
- **Libraries:** : TensorFlow, PyTorch, Keras, Scikit-Learn, Numpy, Jupyter, OpenCV, ROS.

PUBLICATIONS

1. Z. Zhao, Y. Mao, Y. Ding, P. Ren and N. Zheng, "Visual-Based Semantic SLAM with Landmarks for Large-Scale Outdoor Environment," 2019 2nd China Symposium on Cognitive Computing and Hybrid Intelligence (CCHI), Xi'an, China, 2019, pp. 149-154.
2. R. Chen, W. Wang, Z. Zhao, D. Zhao, "Active Learning for Risk-Sensitive Inverse Reinforcement Learning," submitted to ICRA 2020, preprinted in arXiv.

SELECTED EXPERIENCE

Active Risk-sensitive Inverse Reinforcement Learning

Summer Intern in Safe AI Lab, CMU

Pittsburgh, U.S.A.

Jul 2019 - Sep 2019

- **Faculty Advisor:** Ding Zhao, Assistant Professor in Department of Mechanical Engineering & The Robotics Institute, CMU
- **Project Description:** This project is aimed to enable autonomous vehicles knowing the risk situations in dynamic environments by using Inverse Reinforcement Learning. The paper has submitted to IEEE ICRA 2020.

Visual-based Semantic SLAM with Landmarks for large-scale outdoor environments

Research Intern in Institute of Artificial Intelligence and Robotics (IAIR), XJTU

Xi'an, China

Nov 2018 - Present

- **Faculty Advisor:** Pengju Ren, Associate Professor in IAIR, Xi'an Jiaotong University
- **Project Description:** This project built a visual semantic SLAM system with GPS fusion and landmarks association.

Multi-Robot Cooperative Navigation

Research Intern in IAIR, XJTU

Xi'an, China

Apr 2018 - Mar 2019

- **Faculty Advisor:** Pengju Ren, Associate Professor in IAIR, XJTU
- **Project Description:** This project established a multi-robot navigation and exploration system, which consist of UAVs and UGVs.
- **Demo Video:** [Cooperative Navigation System of UAV and UGV.](#)

Tele-Robotics & Deep Learning

Student of 2018 Summer Workshop, School of Computing, NUS

Singapore

Jul 2018 - Aug 2018

- **Faculty Advisor:** Soo Yuen Jien, Associate Professor in School of Computing, NUS
- **Project Description:** We built an autonomous blind-guide robot by using Raspberry Pi and Arduino. We have also actualized the computer vision task py inception model and Azure service for obstacle classification.

SCHOLARSHIPS & HONORABLE TITLES

- **NUS Research Scholarship:** in 2020 - 2024
- **Siyuan Merit Scholarship:** in 2017 & 2018 & 2019
- **Excellent Student:** in 2017 & 2018 & 2019
- **Second Prize of 1989 Mechanical Alumni Scholarship for Qian Class:** in 2018 (4 Candidates out of 119 students in Qian Class)